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First language acquisition, two first languages

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In this chapter, we will discuss how children acquire their native language, paying attention to both the stages in this process as well as the various components of language: sounds, words, and syntax. Why are we interested in knowing the milestones of this process? If we want to argue that the L2A process is a natural acquisition process guided by the innate Language Acquisition Device, it is imperative to compare the two processes at various milestones. We are pretty sure these days that the two processes (monolingual L1A, adult L2A) do not unfold in exactly the same way, but we would still like to know which aspects of the processes may be similar and which may be different. Furthermore, simultaneous acquisition of two languages, or 2L1, is even more pertinent to compare with adult L2A because both conditions make the individual a bilingual.

5.1 Acquisition of the sounds of one's native language

Children need to acquire (a) the sounds of their native language, what linguists call the segmental inventory of that language, (b) the phonological

processes (or rules) that involve these sounds, (c) the restrictions on phonotactics (which sounds may combine in their language and which may not), the word prosodic structure, and larger prosodic units that define the adult grammar.¹ Most importantly, children have to build a lexicon in which phonological representations of words are stored. Two acquisition issues discussed in the literature are the phoneme categorization problem and the segmentation problem.

How do babies know which of the sounds functioning in the languages of the world distinguish words in their own language? After all, there are around 600 consonants and 200 vowels in all human languages (Ladefoged 2001). Each individual language uses a unique set of about 40 distinct elements, or *phonemes*, which distinguish between words. Phonemes actually comprise a number of similar but non-identical sounds, called *phones* or phonetic units, which are functionally equivalent in the language. For example, [l] and [r] belong to different phonemes in English (e.g., *lake* and *rake*), since they differentiate words, but [l] and [r] happen to belong to a single phonemic category in Japanese, the phoneme /r/.² A baby's task is to figure out the composition of the 40-odd phonemic categories of her language. Japanese-learning infants have to group the phonetic units [r] and [l] into a single phonemic category whereas English-learning infants must uphold the distinction in order to separate *rake* from *lake*.

Babies solve this problem by starting big and eliminating options. It turns out that infants are capable of distinguishing all the sounds of the world's languages. Very young infants discriminate not only native, but also non-native phonetic differences (Eimas, Siqueland, Jusczyk, and Vigorito 1971, Streeter 1976), differences that the adult speakers of their native language are not aware of. For example, the [d] sounds in the context of "this doll" versus "our doll" are acoustically different, although they belong to the same phoneme /d/ in English. However, the dental and retroflex variants of [d] belong to different phonemes in Hindi, as they distinguish between the Hindi words for *lentils* and *branch*. Werker and Tees (1984) demonstrated that both English and Hindi-learning infants discriminate this contrast at 6–8 months of age, but only the Hindi-learning infants continue to do so at

¹ Prosody (also known as suprasegmental phonology) includes those aspects of speech that go beyond phonemes and deal with the auditory qualities of sound, e.g., pauses, pitch or intonation, stress or emphasis, volume, and others.

² We generally enclose phonemes in slanted lines and sounds in square brackets.

age 10–12 months. In a series of experiments, Janet Werker and her colleagues showed young babies' sensitivity to various non native sounds and contrasts. Speech perception scientists interpret these findings as indicating that infants are born with special sensitivity to all linguistically important sounds.

But how do scientists know what babies are sensitive to? After all, the infants in these experiments cannot speak, so they cannot tell us about their perceptions. Scientists take advantage of a behavior at which infants are singularly good: sucking. Babies depend on sucking for their sustenance, but they also engage in sucking behavior between feeding, as becomes obvious when you give them a pacifier. During the initial, or habituation, period of an experiment, a machine measures how fast and how often the baby sucks when no stimulus is presented. This establishes a baseline sucking rate and amplitude (how hard she sucks). When a new stimulus is presented aurally, the baby starts sucking with more force and more frequently. When the same stimulus is repeated for some time, however, the baby's sucking gets down to habituation levels, as if signaling that she is "bored" with the old stimulus. If a new stimulus, for example, a new sound, is played at that moment, and the baby notices the change, she is said to dis-habituate, that is, she starts sucking harder again. The whole experimental paradigm is known as the high-amplitude sucking paradigm, or HAS. This paradigm is very widely used in speech perception research. That's how we know that new born babies and infants distinguish various sounds.

In fact, the native phoneme categorization problem is even more complex than I suggested above. When a female speaker pronounces *lake*, the acoustic features of the signal are different from when a male speaker pronounces the same word. Using detailed acoustic analysis, research has established that the phonetic characteristics of any speech sound diverge in the way different speakers pronounce it, and even in the way the same speaker pronounces it at different times. Obviously, some acoustic differences lead infants to phoneme differentiation, while others don't. While infants are sensitive to non native phonemic contrasts, they are not sensitive to acoustic variation that does not lead to phoneme differentiation. Infants may be able to detect a multitude of phonetic contrasts, but by their first year of life they have largely sorted out which contrasts do matter in their language and which don't. In doing this, they have mapped the different variants of a native phoneme onto the same category, while maintaining the differences between phonemes.

Think about whether the phoneme categorization problem applies for L2 speakers as well.

Another all-important acquisition task babies face is the so-called speech segmentation problem: where does one word finish and the other start? Recall the spectrogram of *This is a spoken sentence* in Chapter 3, which had no gaps between words. You may have experienced this problem if you tried to listen to an unfamiliar language on the radio. The problem is particularly acute if that language has different prosodic characteristics, for example, if a Spanish speaker listens to a tone language such as Mandarin Chinese, Vietnamese, or Thai. In fact, this problem persists in native speakers for words that they don't know. For example, a high school teacher, A. Greene, published a collection of word analysis errors that her students made, one of which is the following (cited in O'Grady 2005: 18):

(1) In 1957, Eugene O'Neill won a Pullet Surprise.

If you read this sentence aloud, you would immediately understand that the student was trying to write: *Eugene O'Neill won a Pulitzer Prize*. Not knowing the words, the student came up with a reanalysis of the word boundary.

Teaching relevance: segmentation

Segmentation is an especially difficult problem for L2 learners, too. Think of any examples you have heard from your students, or produced yourself, when you could not distinguish the actual word boundaries. What could a teacher do in order to make segmentation easier for her students?

Is it possible that children learn some words when spoken in isolation, such as *Daddy*? Yes, although this is not going to take babies all the way to solving the segmentation problem. Many of the child's first 50 words are those that the child has heard in isolation. But extensive observation of caretaker speech suggests that only 6 to 60 words produced by a caretaker within an hour are produced in isolation. Think of how that knowledge continues to give in the segmentation enterprise. If you have already isolated *Daddy* as a word, then it follows that the rest of the sequence in the utterance *Daddy knows* should also contain a whole word or words.

What are some other cues that the baby has access to? I already mentioned prosody, and that is indeed a cue that grows in importance after the baby is born. The prosodic bootstrapping hypothesis³ proposes that children pay attention to prosodic characteristics of the speech signal because they can identify word boundaries (although not perfectly). Word stress is a crucial prosodic characteristic. For example, about 90% of disyllabic (two syllable) words in English employ the *trochaic* stress pattern, meaning that the first syllable is stressed, or pronounced louder, while the second is relatively less so. In order to appreciate the pattern, pronounce the words *baby*, *sponsor*, *money* twice: first in the regular way, then make the second syllable louder (and slightly longer). The second way is very unnatural for those words, right? The fact that trochaic words are a great majority in the English language is a significant cue to segmentation: if babies assume that each stressed syllable is the beginning of a new word, they will be right about 90% of the time. Of course, this claim is oversimplifying the matter, but the point is that we have identified two segmentation cues already: words pronounced in isolation, and prosody (stress), which obviously complement each other.

A third cue likely used by older babies to supplement the other cues and cut into the speech signal is *phonotactics*. Phonotactic information is unique to each language, and it comprises the possible combinations of sounds in that language for each syllable position (*onset*, *nucleus*, and *coda*, or the beginning, middle, and end of each syllable). For example, in English a combination of the consonants [t] and [l] cannot start a syllable, so the sequence *fatlump* has to be divided into *fat lump* and never *fa thump*. However, in Bulgarian this combination of sounds is allowed in the beginning of words, hence of syllables, (e.g., [tləst] ‘fat’), so a Bulgarian-learning baby cannot safely decide that the syllable or word boundary is between [t] and [l]. Another example would be the segmentation of the phrase *thisplace*: it can only be segmented into *thisp lace* or *this place* but not into *thispl ace*. Think of other words that finish in *-isp* and *-is* in English: you will never be able to come up with a word that finishes in *-ispl* because this coda does not obey English phonotactic constraints.

³ The name of this hypothesis (and others, as we shall see later) come from the expression “to pull yourself up by your bootstraps,” meaning, to help yourself. A famous literary character, Baron Münchhausen, attempted to do just that, when trying to go to the moon.

Production of native sounds and words is another matter altogether. It is well known that children's first words approximate but do not match adult productions. Research into child production has argued that children gradually build up a system of phonological contrasts, phonotactics, and prosodic structure, describing typical processes like segment deletion, substitution, consonant harmony, and metathesis (reordering the sounds in a word). Exhibiting substitution, Daniel, an English-speaking child, produced [gak] for *clock*, *sock*, *rock*, *quack*, and he used metathesis (switched sounds around) when he said [deks] instead of *desk*. An intriguing research question is whether children's phonological system of representations differs from that of adults. The fact that they recognize words that they cannot pronounce suggests that there is continuity between child and adult representational systems. Since adult L2 speakers do not have production problems that are commensurate with babies', however, we shall leave this issue aside.

To recapitulate, babies acquire segmental inventories and knowledge about phonotactic and prosodic structure largely in the absence of a lexicon. In fact, that knowledge *has* to be acquired before lexical learning starts, since it guides word learning. It is likely that children use all the cues available to them (prosody, phonotactics, distributional cues, etc.) in conjunction, so that the various cues can reinforce each other.

5.2 Learning word meanings

Children produce their first word around the first year of their life, give or take a couple of months. It takes them about six to eight months to get to 50 words, that is, at first, word learning is quite slow. However, after that age, something called "the great vocabulary spurt" takes place: a time when children learn one or two new words a day (McMurray 2007). At later ages, children learn even faster: about ten words a day at ages 2 to 6 years. The average six-year-old has 14 000 words in her mental lexicon, while the average high school graduate knows about 60 000 words (Bloom 2000). Compare this amazing rate of learning to word learning in L2 acquisition, which is typically much slower. Not only the rate of acquisition is amazing: children can deduce the meaning of a word from hearing it used a couple of times; older children can learn a word from one occurrence in speech. We call that *fast mapping*.

This is an astonishing feat of deduction on another account, too: the child has to overcome a significant Poverty of the Stimulus. The same Poverty of the Stimulus is to some extent present for L2 learners in communicative classrooms, and especially true for naturalistic ones. This problem of acquisition is discussed wittily and convincingly by the philosopher Quine who dubbed it the Gavagai Problem (Quine 1960). Imagine that a child is in the company of an adult caregiver, and they see a bunny rabbit running away in the distance. The adult says “Gavagai.” What is the child to assume as the meaning of the word *gavagai*? It can mean “rabbit”, but it can also mean “running”, or “running away from us” or “furry animal” or “two ears” or “let’s eat him.” The information on the object to be named, provided by the extralinguistic situation does not allow the learner, whether child or adult, to identify one and only one possible meaning.⁴ Children employ several cognitive strategies to resolve this word-learning problem. As in the speech segmentation problem, children entertain hypotheses, but not all logically possible hypotheses.

The first strategy that can help children with the Gavagai Problem is the so-called *whole object assumption*. It proposes that children who are exposed to an object and a name will assume that the name stands for the whole object and not for parts of it. That means that they will not assume *gavagai* names the bunny’s tail or legs, but the whole bunny. In general, infants as well as adults have a natural predisposition to treat novel words as labeling the whole object (Markman and Hutchison 1984). Of course, this assumption has to be overruled when the words for body parts, such as *leg* and *fur*, need to be learned.

Next, there is *the type assumption* (Clark 1993). In order to exclude individual names such as Flopsy or Jack, children should assume that the word labels a whole type of similar animals, not one individual rabbit. Finally, another well-studied assumption is the *basic level assumption* (Hall and Waxman 1993). It postulates that labels of objects will be assumed to refer to basic-level terms (e.g., PERSON) not a context-restricted kind (e.g., PASSENGER). A new count noun applied to an unfamiliar solid object should refer to a basic-level kind of object and not to a kind that distinguishes its members based on the stage or situation they are in.

⁴ Recall that this learning situation is called “poverty of the stimulus.” We discussed it in Chapter 2.

Another learning assumption that saves a lot of effort when it comes to labels of objects is the *mutual exclusivity assumption* (Clark 2003, Markman and Wachtel 1988). Proposed by Ellen Markman, it postulates that no two words in a language should have exactly the same meaning. O’Grady (2005) rightly calls this an organizational assumption. In one experiment, children were shown a familiar object, e.g., a spoon and an unfamiliar object, e.g., a whisk. Then a researcher called the unknown object *a fendle* (a made-up word obeying English phonotactic rules) and asked the children to bring the fendle over to them. The children brought the unknown object to the researcher. They reasoned that they knew what a spoon was, there were two objects around, and they had to get the fendle. If one of the objects was a spoon, then the fendle had to be the other object, since a spoon could not be called a *spoon* and *fendle* at the same time.

The mutual exclusivity assumption works well for children with small vocabularies. It helps children to override the whole object assumption. In one possible scenario, if a child already knows the word *dog*, and the adult who is sharing a book with the child points to a picture of a fluffy dog, saying “fluffy”, by mutual exclusivity, the child should assume that the new word is not naming the object again, but is referring to a property of the dog. That would be the correct assumption.

Acquiring the meanings of verbs present an even bigger challenge than nouns. First of all, consider that verbs are cognitively much more complex than nouns, especially the first nouns in the child vocabulary, which mostly describe objects. Knowledge of verbs crucially includes knowing the number of participants in the verb event. For instance, knowing the verb *eat* includes the thematic roles of the two participants in an eating event: one of the participants (the Agent) has to be a sentient being (a person, a rabbit) and the other participant (the Theme) has to be edible. *Charlie ate his shoe* is not ungrammatical but pragmatically odd because shoes are not normally edible items. However, recall the film *The Gold Rush* (1925), in which Charlie Chaplin plays a tramp who eats his own shoe.

Linguists debate whether knowing the meaning of the verb gives rise to correct sentence syntax or whether knowledge of the syntactic frames in which a verb appears “bootstraps” children into acquiring verbal meanings. In other words, this is a debate on whether lexical semantics comes first and syntax follows, or the other way around. These two opposing views are called “semantic bootstrapping” and “syntactic bootstrapping,” respectively. To understand syntactic bootstrapping, we need to know what is

meant by syntactic frames. If you hear a novel verb in a sentence where you know the rest of the words, chances are you will be able to guess at its meaning. Take the sentence in (2):

- (2) Mary *clessed* the toy car to her baby brother.

The made-up verb *cless* has three arguments in this sentence: Mary, the toy car, her baby brother. A plausible assumption will be that the sentence denotes a (past) verbal action akin to acting upon (such as showing or passing) the inanimate object (the car) to bring it to the attention or possession of the animate participant (the brother). That is how the verbal frame Noun–Verb–Noun–to Noun might bootstrap acquisition of *cless*, if it were a real verb. In a sense, the speaker translates the syntactic frame into the semantic frame Agent–Verb–Theme–Recipient. Undoubtedly, transitive frames (“Mary *clessed* the toy car”) present more difficulty, hence lead to more variable hypotheses about verbal meaning.

The syntactic frame Noun–Verb–Noun happens to be mapped onto Agent–Verb–Theme (where the Theme is the affected participant in the verbal action) around 75% of the time in English child-directed speech (Kline and Demuth 2013). No wonder young children sometimes overextend this causative meaning to intransitive verbs as in (3) (Bowerman 1974):

- (3) a. Don’t giggle me. (a child at age 3;0)
b. I’m singing him. (another child at 3;1)

It has been established that frequencies of occurrence of each syntactic frame have an effect on comprehension and production of novel verbs in sentences. As we mentioned above, the most frequent frame is the transitive one as in (4). Next comes the Agent–Intransitive Verb frame as in (5) with 16% frequency, followed by the Theme–Intransitive Verb frame as in (6) with 9%.

- (4) Joey is closing the door.
(5) Joey is walking.
(6) The door is closing.

Accuracy with verb meanings follows these frequency distributions (Kline and Demuth 2013). At age 3, children show remarkable command of the

syntactic frames by producing intransitive sentences that respect the event restrictions (who is doing what) and animacy cues (the affected participant (the Theme) is likely to be inanimate). They demonstrate comprehension of verb frames even at the age of 2. What is debated at present is whether children achieve that accuracy learning verb-by-verb (the gradual development view) or rapidly, generalizing across syntactic frames (the early generalization view). In sum, word learning is at the heart of language acquisition, and many of the psychological processes involved in monolingual acquisition are relevant for second language acquisition as well.

5.3 Acquisition of functional morphology and syntax

Examining verb learning brings us to the next area of grammar building, that of morphosyntactic development. When we combine words in some order, that order is crucial for understanding the sentence, at least in English and other languages. But the word order is not the whole story in attributing the sentence a representation that leads to correct comprehension. Speakers also need to pay attention to the endings of words (*-ing*, *-ed*) or the little words (auxiliaries, definite articles, prepositions, etc.) that signal the grammatical meaning (see Chapter 3). Language researchers pool these grammatical morphemes under the label *functional morphology*. To use whole sentences in communication, children have to learn to decode what the order of the words entails for the meaning of the whole sentence (in comprehension); they have to learn to line up their words in the appropriate order (in production), as well as decode/signal grammatical meaning.

At this point, it is worth emphasizing the difference between lexical and grammatical meaning. In this chapter, children's early knowledge of nouns and verbs was assumed to suggest they know the *lexical meaning* of the items, that is the idiosyncratic denotation of each lexical item such as "chair" denotes a **PIECE OF FURNITURE USED FOR SITTING**, "push" denotes an **ACTION OF MOVING AN OBJECT ALONG A SURFACE**, etc. Children do learn lexical meanings one by one but may also generalize meanings to whole classes of words. *Grammatical meanings* are of another order altogether. They reflect cognitive universal concepts (time, space) that are also linguistically relevant in the sense that languages of the world tend to mark them in their structure. Examples of such concepts include [plural noun], marked by the *-s* affix and [past state or

event] signaled by *-ed* in English. It is also important to distinguish between grammatical meanings and their *linguistic expressions*. For example, the plural meaning is expressed by reduplication in Indonesian and Malay, or left unexpressed morphologically in Acehnese. The past meaning is not signaled by dedicated inflectional morphology in Mandarin Chinese, Vietnamese, and Thai. The World Atlas of Language Structures <http://wals.info/> is a precious online resource compiling linguistic research on grammatical meanings that you can explore at your leisure. See the map for past tense distribution here <http://wals.info/feature/66A#2/25.5/148.4>.

Children are very good at acquiring regular morphological inflections, such as the plural *-s*. This is demonstrated in a classic experiment performed by Jean Berko Gleason in 1958. Young children were shown a drawing of a made-up creature, described by the pseudoword (or nonce word) “wug,” in this way. In this case, the creature looked somewhat like a bird and was blue. Then another creature of the same shape appeared, and children had to produce the correct word ending. You can see a historical footage from this test in this YouTube video <https://www.youtube.com/watch?v=MgB2iMuEZAA>.⁵ Preschoolers at the age of 4 and 5 accurately produced the suffix 75% of the time.⁶ This high accuracy on some, though not all, novel tokens represents children’s ability to formulate a grammatical rule: when there is more than one noun, add an *-s*. The Wug test is probably the most famous experiment in language acquisition research.

Rule-learning is implicated in acquiring another inflectional suffix, past tense morphology in English. (Gleason tested it in the Wug test, along with the plural and other affixes.) Recall the Great Past Tense Debate we discussed in Chapter 3. Even though the majority of verbs create a past tense form by adding the regular *-ed*, there are irregular verbs whose past tense forms are marked in other ways: *eat–ate*, *sing–sang*, *go–went*, etc. Children’s behavior with these forms is indicative of how learning of functional morphology involves both token learning and generalized rule learning. It can be modeled by an idealized U-shaped learning curve. At first, the child produces a few tokens of the correct irregular form of frequent verbs,

⁵ You can see the whole experiment at the following blog: <http://www.onbeing.org/blog/sunday-morning-exercise-take-wug-test/2510>

⁶ They were less accurate when the pseudowords ended in a sibilant sound such as *kash* and the correct ending needed a vowel in front of the *-s*: *kashes*, pronounced [kaʃɪz]. See Berko Gleason (1958).

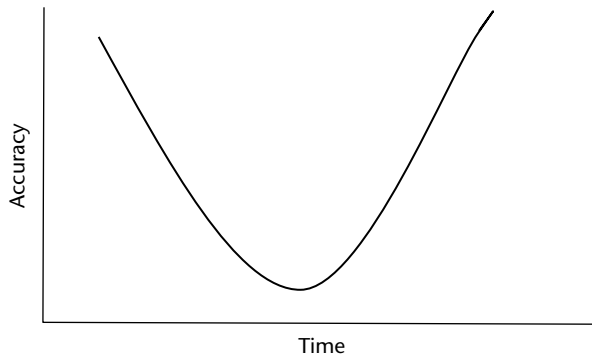


Figure 5.1 Idealized illustration of U-shaped development

say *went*, *had*, *took*, *gave*. Then she learns a lot of regular verbs (*looked*, *dropped*), figures out the rule and behaves as if she has unlearned the correct irregular form by over-applying the new rule. At this stage she produces *goed* and *went* interchangeably. Finally, she relearns the correct irregular forms and keeps them separate from the regular forms. Plotting the accuracy on verb past tense forms against the number of verbs learned, we get the following idealized shape of the letter U (Figure 5.1).

However, this U-shaped behavior is not uniform, in the sense that it does not affect all the irregular verbs at the same time (Marcus et al. 1992). That is, there are phases of over-regularization for different verbs at different times. Frequent verbs are over-regularized much less often than infrequent ones. There are indications that the past tense is not the only overgeneralized affix in English. Children also over generalize the plural suffix as in **mans*, **tooths* and the comparative and superlative affixes of comparatives: **gooder*, **powerfullest*, among others. They can certainly find regularity where it doesn't exist, as in the following exchange. Explain which affix is overgeneralized where.

- (7) a. Parent: No booze in the house!
 Child: What's a "boo"?
 b. [After being told it was going to snow.]
 Child: "It did! It snow!"

As we already mentioned in Chapter 3, linguists and psychologists have been debating how best to explain this sort of behavior. One account proposes a dual lexical access mechanism, where regular and irregular

verbs are retrieved in different ways (Marcus et al. 1992, Pinker and Prince 1988). On the one hand, children memorize past marking on separate lexical items; on the other hand, they learn a rule of regular past tense inflection. Another account argues for a single lexical access mechanism and contends that U-shaped development is due to changes in vocabulary size (Plunkett and Marchman 1991, 1993). The mental representation of verbs is a set of connections in a network instead of rules, and learning involves forming associations between sound sequences and meanings. Eventually the correct form wins out by establishing the strongest connection. It is during this period of competition that overregularization occurs because the connections for the irregular verbs are still not strong enough to always overtake the regular morpheme. Whatever explanation turns out to be correct, the process of (unconscious) overgeneralization of inflectional morphology is a fact of learning language, and we will see it exemplified in SLA learning as well.

It is worth making a conscious detour from functional morphology at this time. When we pondered the distinction above between grammatical meaning and its expressions, we mentioned the need to keep in mind that grammatical meanings are universal, while their expressions are language specific. If you access the WALS map of tense, you will see that roughly half of the languages studied do not mark tense, the morphological expression of time (the white circles). The fact that Mandarin Chinese, for example, uses no past tense morphology does not mean that Chinese speakers do not express temporality, i.e., cannot situate an event or state on the time line. They signal temporality with other, non-morphological linguistic means, such as context, time adverbials, and aspectual morphemes (Smith and Erbaugh 2005). So when we discuss the acquisition of functional morphology, we have to acknowledge that very often we are examining the acquisition of grammatical meanings and their various expressions in a specific language, which might not always be functional morphemes.

Let's take the following learning situation as an example. Children acquiring Chinese have a different problem compared to children acquiring English. Acquiring grammatical meanings in the absence of dedicated morphology may be even harder than acquiring tense markers, because children are forced to identify other markers of temporality, which come from various areas of the grammar: adverbials in the lexicon, grammatical morphemes reflecting other grammatical meanings, and pragmatic principles. To master the temporal system, Mandarin-speaking children need

to learn not only how to use formal temporal devices (the aspectual particles, for example) but also how to convey implicit references to past or future events. In her pioneering research on this issue, Erbaugh (1992) suggests that learning “less” is actually more difficult. There are indications that learning to signal temporality is not yet accomplished by Mandarin-learning children at the age of 3;2 (Huang 2003). Children’s abilities to use discourse-pragmatic resources may still be rather limited at that age. That is why they rely on situational context (the here and now) and not so much on discourse context (previously mentioned temporal adverbials) in their marking of temporality. End of detour.

The idea that, in learning morphosyntax, children are building a mental grammatical system came to the forefront of language acquisition studies in the 70s. The towering figure of this line of research is the psychologist Roger Brown, with his description of longitudinal data from the linguistic behavior of three children (who he called Adam, Eve, and Sarah). The children’s speech was followed over a period of 15 to 21 months, starting at roughly 2 years of age. Brown (1973) observed that the children acquired 14 grammatical morphemes, which he called functors, in roughly the same chronological order.⁷ DeVilliers and DeVilliers (1973) confirmed the findings with cross-sectional data. The seminal idea was that all children acquire their first language in a fixed, universal order, regardless of the specific grammatical structure of the language they learn.⁸ The progressive *-ing* morpheme was acquired first, then came prepositions *in* and *on*, followed by regular plural *-s*, and so on. Brown argued that this order reflected the cumulative grammatical complexity of the morphemes, measured in the number of features and sometimes, depending on the grammatical theory, in the number of rules they employ. See Table 3 in the exercises of this

⁷ Brown (1973) defines functors as “forms that do not, in any simple way, make reference. They mark grammatical structures and carry subtle modulatory meanings. The word classes or parts of speech involved (inflections, auxiliary verbs, articles, prepositions, and conjunctions) all have few members and do not readily admit new members” (p. 75). Is he defining what we discussed as morphemes with grammatical meaning?

⁸ In his APA address, Brown summed up his view of language development in a sentence: “All of this, of course, gives a very ‘biological’ impression, almost as if semantic cells of a finite set of types were dividing and combining and then redividing and recombining in ways common to the species.” (Cited from Roger Brown’s obituary by Steven Pinker.)

chapter for the full list of functor morphemes. In later years, researchers explored other factors to explain this order of acquisition, for example, the frequency of these morphemes in child-directed speech as well as the phonetic salience of the morphemes (how easily they are perceived in the speech signal). Although there is no consensus on the issue, it is clear that the acquisition order may be due to a variety of factors.

To add a typological perspective to the acquisition of inflectional morphology, we should briefly consider the instructive research of Wolfgang Dressler's research group (Laaha and Gillis 2007). The researchers compared the rate of acquisition of verbal and nominal inflectional paradigms in nine languages that they categorized as weakly inflecting, strongly inflecting, and agglutinating (three languages in each category). Agglutinating languages such as Turkish, Finnish, and Yucatec generally express one meaning per inflectional morpheme. In "strongly inflectional" languages such as Russian, Greek, and Croatian, in contrast, one morpheme can carry several grammatical meanings. "Weakly inflectional" languages (French, German, Dutch) mark fewer meanings with inflectional morphology. The researchers computed the mean paradigm size of these groups of languages and correlated this size to child acquisition data of the same morphology. The overall conclusion was that the richer noun or verb morphology in the input, the more rapidly the child developed that morphology. It is as if the child is rising up to whatever challenge the morphosyntax of her language throws at her.

Children's syntax grows rapidly in their third year of life. As demonstrated already in Brown's and much later work, the vast majority of the sequences of words they produce obey the word order of the language they are learning. Researchers of child language usually distinguish a *one-word stage*, a *two-word stage*, and a *multiple-word stage* in children's syntactic development. Children's early utterances represent snippets from longer potential sentences expressing a more complicated proposition. For example, most two-word sequences contain a verb and one argument, but that argument can have various thematic roles, e.g., *Mommy fix* (Agent–Verb), *give doggie* (Verb–Recipient), and *put light* (Verb–Location). The complex proposition that children are able to express later is probably limited at the two-word stage by a processing bottleneck.

One widely used measure of children's syntactic development is mean length of utterance (MLU). It is a way to monitor syntactic growth over time. This is accomplished by counting the number of morphemes (both

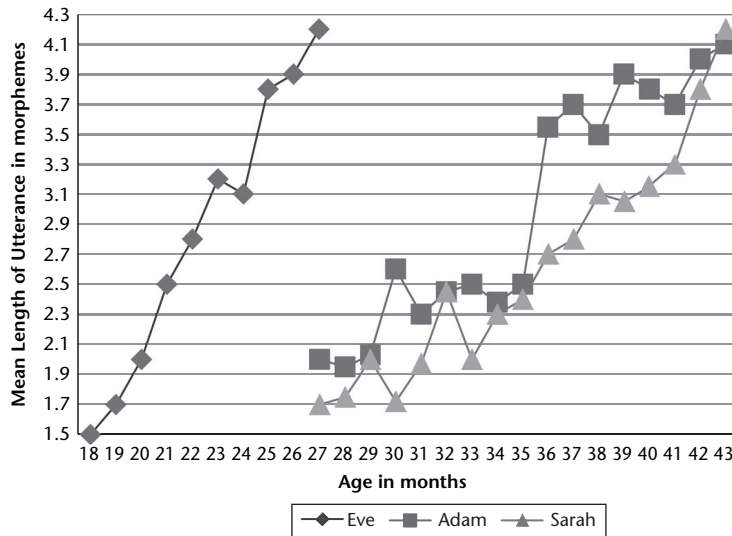


Figure 5.2 Mean length of utterance and chronological age for the three children from the Brown corpus, following Brown (1973)

lexical and grammatical morphemes) in a set of 100 sentences, and then dividing by 100. You will have the opportunity to try this for yourselves at the end of this chapter. Figure 5.2 following Brown (1973) illustrates the MLU development in Eve, Adam, and Sarah.

As you can ascertain visually, Eve's syntactic development is quite rapid, and she reaches MLU of over four in the course of eight months, while the other children's syntactic spurt starts a bit later and lasts a little longer.

Since we cannot do justice to the vast topic of syntactic development here, I shall exemplify more complex structures in child language with *wh*-question formation. The first patterns that appear in child production, (e.g., *What's this? What's that? What this? Who that?*) suggest that children use these word sequences as unanalyzed chunks, as sequences that do not have internal structure. However, it is also notable that English children produce spontaneous *wh*-initial strings (e.g., *where go? what hit?*) that they cannot be repeating. (Think about why not?) While *wh*-words appear sentence-initially in child English, they appear sentence-internally in child Indonesian, a language where *wh*-words appear *in situ* (Cole et al. 2001). This distinction suggests that the *wh*-movement parameter, which regulates whether *wh*-words move to the beginning of the sentence or not, is set rather early.

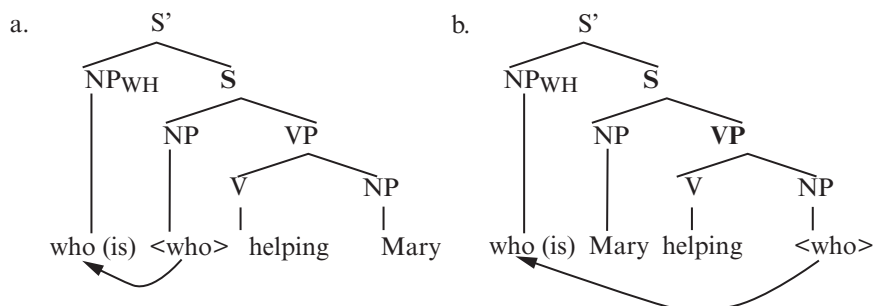
Stromswold (1995) examined the spontaneous speech of 12 English-learning children from the CHILDES database, in order to establish whether subject or object questions appeared first. The results seem to be mixed (O'Grady 1997), without clear preference for one pattern over and above what can be explained by frequency in parental speech. However, in forced-production experimental tasks, clear preference for subject questions emerged: children made fewer errors in subject questions (Wilhelm and Hanna 1992, Yoshinaga 1996). This has also been found for languages such as French, Italian, and Hebrew. Why is this important? Because it sheds light on the child's process of building a mental grammar.

In trying to explain children's superior accuracy on subject questions over object questions, O'Grady (1997: 135) notes that subject questions do not, at least on the surface, exhibit movement of the auxiliary over the subject (subject-auxiliary inversion):

(8) Who is helping Mary?
subject-aux-verb-object

(9) Who is Mary helping?
object-aux-subject-verb

(10)



This subject-object asymmetry cannot be due solely to frequency in parental speech, as ascertained by Stromswold (1995). A structural reason suggested by O'Grady (1997: 136) is the added complexity of the object questions. He formulates it as follows:

(11) "A structure's complexity increases with the number of XP categories (S, VP, etc.) between a gap and the element with which it is associated."

Recall that we mentioned structural complexity while discussing Gibson's Dependency Locality Theory in Chapter 3. This is essentially the same explanation. Let us look at the approximate structures in (10a) and (10b), corresponding to sentences in (8) and (9). S stands for Sentence and the node above S is S' (S-bar), a higher functional category where *wh*-phrases land. Angled brackets < > mark the original position of the *wh*-word, subject or object of *helping*. The position from which movement originates is also known as "the gap." In moving to S', the subject *who* crosses only one category, S. To land in the same position known as "the filler," the object *who* has to cross at least two categories, VP and S. This larger number of crossed nodes makes the object questions structurally more complex, hence more difficult for children.

If this structural complexity explanation is on the right track, it makes predictions about child language: the longer the structural distance between filler and gap, the more difficulty children will experience. Hildebrand (1987) used an imitation task to check this prediction. In this task, children are invited to repeat a sentence verbatim. It is considered that they can correctly repeat a sentence only if it is allowed by their grammar and can be correctly parsed. Some of Hildebrand's test sentences appear below; Table 5.1 shows the percentage of correctly repeated sentences in the task. Keep in mind that the words in the angled brackets are not pronounced; these are the original positions of the moved *wh*-words, or the gap. Square brackets stand for nodes that have to be crossed in *wh*-movement to the filler; those are additionally marked underneath with a number sign (#).

(12) What did [_S the little girl [_{VP} hit <what> with the block today?]]
#

(13) What did [_S the little boy [_{VP} play [_{PP} with <what> behind his mother?]]]
#

(14) What did [_S the boy [_{VP} read [_{NP} a story [_{PP} about <what> this morning?]]]]
#

Table 5.1 Percentage of correctly repeated sentences in Imitation Task, Hildebrand (1984). Presented following O'Grady 1997: 137

Example sentence	4-year-olds	6-year-olds	8-year-olds	10-year-olds
(12)	83	94	97	100
(13)	46	80	86	97
(14)	31	78	89	89

Conspicuously, the majority of the errors involved children reducing the structural complexity of the sentence they had to repeat: e.g., *What did he play with?* was repeated as *What did he play?* These results have been corroborated with other experimental techniques in comprehension as well as production.

Let us ponder the issue of competence versus performance in this case. If you need to, review these concepts introduced in Chapter 1 and the structural complexity discussion in Chapter 3. After the initial stage of repeating unanalyzed strings, children may acquire the way English creates *wh*-questions (the *wh*-word moves to sentence initial position, there is subject-auxiliary inversion⁹) in stages. It makes sense to assume that children acquire a general rule of grammatical feature-based operations, and not separate rules for each kind of question (subject, object, object of a preposition, adjunct). If this is the right explanation, the structural complexity proposal by O'Grady and Gibson can manifest itself mainly in processing: children have underlying knowledge of both types of questions but object questions are more difficult to process, leading to more errors. Surveying a large body of data, Roeper and de Villiers (2011: 239) come to the similar conclusion that length of derivation is “both guiding and constraining acquisition.” In other words, children know the rules for how to create questions, they just make more errors in performance with the more complex sentences.

Teaching relevance: structural complexity

Do you think that structural complexity will affect adult second language acquisition? Why or why not? Is it possible to ameliorate structural complexity so as to aid learners at initial stages in acquisition? These questions can all be answered in the affirmative. If teachers know what is structurally complex (from personal experience and from linguistic theory), then they can break down the acquisition process into manageable steps by reducing complexity.

⁹ Although we have not discussed subject–auxiliary inversion in this chapter, the name illustrates the process: the subject and the finite auxiliary switch places in question formation. This type of inversion is illustrated with the statement *We should buy apples* and the corresponding question *What should we buy?*

Finally, let us consider a frequently made claim: normally developing children acquire (the morphosyntax of) their native language by the time they are four years of age, thereby entitling them to be called “linguistic geniuses” who have “cracked the language code” (Pinker 1994). This is a curiously widespread myth, but a myth nonetheless. To demonstrate that, I shall continue with examples from *wh*-acquisition. A recent experimental study on specifically language-impaired children, Van der Lely, Jones and Marshall (2011), included three groups of typically developing children at the ages of 6, 7, and 8. They were tested with three standardized tests to ensure they fell into the normal range of grammatical and verbal abilities. All the children were given sufficient context and were asked to judge the grammaticality of simple *wh*-questions such as *Who kicked Cookie Monster?* and *Who did Cookie Monster kick?* The ungrammatical test sentences included a no-tense error (*Who kiss Miss Piggy?*), doubly marked tense (*Who did kicked Cookie Monster?*) and no *do*-support (*Who Tinky Winky tickled?*). All the groups were highly accurate in accepting the grammatical sentences and semantic control sentences. The accuracy of the children in rejecting the ungrammatical sentences appears in Table 5.2.

Can you comment on the performance of the 6-year-old participants? Does performance reach ceiling in the older participants? Van der Lely et al. observe that only four (36%) of the youngest children performed statistically differently from chance in rejecting these errors, while even among the oldest participants, 2 out of 12 children failed to reject the badly formed sentences. These are hardly isolated results; they surface in study after study. The conclusion is that successful acquisition of syntax is hardly accomplished at 4, and may continue until 7 or 8, depending on different properties.

Table 5.2 Percentage of correctly rejected sentences in Grammaticality Judgment Task, selected from van der Lely et al. (2011)

Type of error	6-year-olds	7-year-olds	8-year-olds
No tense	63.6	76.7	78.5
Double tense	55	85	87.7
No <i>do</i> -support	50	73.4	81.5

5.4 Acquisition of semantics and pragmatics

If there are complex structures that take more time to master, are there some grammatical meanings that are more difficult than others? Research has uncovered an interesting disconnect between children's fairly quick and accurate acquisition of some functional morphology and their knowledge of the actual meaning of that same morphology. For example, in several experiments checking the interpretation attributed to the Imperfect tense in Italian, the imperfective aspect in Polish and the Imperfect and Progressive tenses in Dutch, Angeliek van Hout established that children have interpretations curiously diverging from the adult interpretations, even though they produce these morphemes fairly early. These are all expressions of the grammatical category of viewpoint aspect, where speakers can choose to present the verbal action as ongoing at some past moment and incomplete (unfinished), but not as completed. Children in the age range of 3 to 5 allowed (correctly) imperfective forms to refer to incomplete or ongoing events but also to completed events. Adults did not allow the latter interpretation. Van Hout (2005) argues that what is missing in the child grammar at this stage of development is how to anchor the described events in the time sequence of the larger discourse. Kazanina and Phillips (2007) demonstrated the same comprehension problems with Russian children's performance (aged 3–6) on interpreting imperfective morphology. The study also shows that the children's performance improves dramatically when they are given an explicit temporal clause with respect to which to interpret the incomplete, or ongoing event (e.g., *While the boy was watering flowers, the girl was cleaning the table.*) It seems that assuming this temporal perspective without verbal support is not impossible, but just very hard for the developing mind.

The findings on the acquisition of another type of meaning are relevant here, the so-called scalar implicatures. Sentences with scalar implicatures refer to meanings that are not overtly expressed but just implied. Let's illustrate this with an experimental situation. In a study of Greek-speaking children, Papafragou and Musolino (2003), the experimenters acted out with toys the following situation in front of their subjects. A number of horses played in a meadow, then one by one they jumped over a white fence. At the end, all the horses were on the other side of the fence, having jumped over it. The reasonable expectation is to describe the situation with a sentence like (15):

(15) All the horses jumped over the fence.

However, the researchers had a puppet describe the situation with (16):

(16) Some of the horses jumped over the fence.

Note that the utterance in (16) is logically true: if all the horses jumped over the fence, then some of them jumped over the fence, too. However, the utterance in (16) is not maximally informative. The quantifiers *<some, all>* constitute an implication scale, where the use of the weaker member, *some*, implies the negation of the stronger member, *not all*.¹⁰ In other words, the description in (16) is logically true but pragmatically infelicitous. Greek adults rejected this infelicitous description with 92.5% accuracy. However, Greek 5-year-olds rejected it only 12.5% of the time, a significant adult–child discrepancy. It is as if children do not “take into account” the other, more felicitous description of the same event in (15), which helps adults reject (16) as infelicitous, or not informative enough.

These two hurdles in the child language acquisition of meaning, the interpretation of imperfective grammatical morphology and scalar implicatures, suggest that not all meanings are created equal, even if they belong to a universal inventory of grammatical meanings that all languages have to express in one way or another. Some researchers have argued that cognitive maturity is a factor in developing these and other interpretations: children up to puberty are just not mature enough to take a perspective different from their own in the here-and-now. They have difficulty seeing the world through the eyes of another speaker. Other researchers (e.g., Guasti et al. 2005, Reinhart 2006) point to successful acquisition when the different perspective is made explicit, and argue that in fact processing considerations come into play. Comparing the child’s own perspective with some other person’s perspective is cognitively more demanding, since one has to keep both in working memory for the extent of the comparison. This type of

¹⁰ Sentences like (16) are said to be violating Grice’s (1968) Maxim of Quantity: the speaker should strive to be as informative as is required. The hearer assumes that the speaker has been maximally informative, and if the speaker has not produced the stronger scalar term, then the weaker scalar term is invoked. Note the involvement of two perspectives in this meaning computation: the speaker’s and the hearer’s.

computation uses up a lot of processing resources. When children's processing power proves unequal to the task, children often resort to guessing.

Analogous to structural complexity, Reinhart (2006) formulates this issue as one of *computational complexity*, and I will return to this issue in later chapters. Note that computational complexity does not completely overlap with derivational complexity demonstrated in the case of forming *wh*-questions. Scalar implicatures appear in very simple sentences. In semantics and pragmatics, computational complexity refers to comparing two possible construals and choosing the appropriate or relevant one among them. It is still processing complexity in the sense that the language processor and working memory resources are involved, but of a slightly different type.

Concluding the sections on child language, let us recount the challenges in language development we have mentioned. Children have to break into the speech signal, and they do this making use of various cues. Children learn words at an amazing rate and utilize various strategies to help organize the items in their mental lexicon into word classes (verbs, nouns, etc.) as well as into similar types within each class (transitive verbs, causal verbs, etc.). Acquisition of functional morphology is fast and relatively error-free. The bigger the inflectional paradigm in a language, the faster it is acquired. Children are not only fast but also efficient in extrapolating grammatical rules from comprehensible speech. Complex structures (long *wh*-questions, relative clauses) are not produced or comprehended accurately until later ages (5–8 years), probably because they are difficult to process. They also may not be evidenced in the comprehensible input as often as earlier acquired structures. Since the acquisition of meaning depends on observation of the extralinguistic situation in which a sentence is appropriate, some subtle meanings, not evidenced frequently, also take many years to develop. Akin to structural complexity in syntax, we invoked the concept of computational complexity in the acquisition of semantic and pragmatic interpretations.

5.5 Bilingual first language acquisition

Children who grow up acquiring two languages from birth are known as simultaneous bilinguals. I mentioned quite a few studies involving simultaneous bilinguals in Chapter 4, Section 4.5. In this section, I will review the central issues that have been discussed in the literature on this topic:

Namely, does the child grow up with two differentiated linguistic systems, and if so, do these two systems influence each other later in life? The main point I will try to make, following the bulk of the literature on cross-linguistic influence in bilinguals, is that if language A influences language B, this happens in highly constrained and linguistically motivated ways.

But let me start with the basic question: Is it possible to acquire two languages simultaneously? The research in the last 30 years suggests that the answer is yes. Starting with the seminal work of Genesee (1989), Meisel (1989), and De Houwer (1990), the evidence on morphosyntactic acquisition demonstrates that bilingual children acquire them as largely independent systems. Three main arguments are presented in support of this view: (i) bilingual babies separate their languages from early on, (ii) bilingual children pass through the same developmental stages as monolingual children (sometimes more slowly or more quickly), and (iii) bilingual children commit the same types of error as monolingual children. In addition, as Hoff et al. (2012) demonstrates, in children exposed to two languages, the rate of development of each language will vary as a function of the children's relative amount of exposure.

Is it possible to acquire two languages to monolingual standards, if you start from birth or soon after? That is a different story altogether, and I already touched upon this topic in Chapter 4, arguing that the monolingual yardstick is probably not the best one with which to measure bilingual development. This is because it is hard to imagine that the two languages of a simultaneous bilingual do not interact and influence each other. For this to be true, bilingual children should have had two brains, not two languages in one brain. A major impetus to this line of investigation has been the work of Müller and Hulk (2001). These researchers proposed that cross-linguistic influence occurs in areas of the grammar that are difficult for monolingual children, as well, such as the interface between the syntax and discourse.¹¹ They also suggested that interference may occur where the two languages overlap on the surface. If a structure in language A can be analyzed in two different ways, that is, if it is structurally ambiguous, and language B has evidence for one of the two possible analyses, that analysis would be reinforced in language A. For example, if a child is acquiring a null-subject language (say Greek or Italian) and a non-null-subject language (Dutch,

¹¹ There is quite a lot of evidence suggesting that cross-linguistic influence occurs in other domains, too.

English), this hypothesis would predict oversuppliance of overt subjects in the null-subject language.

Can you discuss the different logical steps of this prediction?

More recent research on the effects of having two grammars has brought other explanatory factors to the limelight: choosing the most economical way to derive a complex structure. Strik and Pérez-Leroux (2011) makes use of this notion of economy to explain why children learning Dutch and French leave their Dutch *wh*-words in their argument position (*in situ*, as linguists say), an option not allowed by that language. French–Dutch bilingual children sometimes produce sentences like (17) instead of the target one with the *wh*-word *what* moved to the front of the sentence and with the subject and verb inverted.

- (17) Jij doe wat giraffe?
 you do.2SG what giraffe
 ‘What are you doing giraffe?’

Even when they do move the *wh*-word to the beginning of the sentence, they sometimes fail to invert the subject and the verb. We already saw that constructions can be compared based on how many categories a *wh*-word crosses over in its movement. A similar notion is that constructions can be compared based on how many Merge operations are involved in the computation (Jacubowicz 2005). The less complex construction, involving fewer terms, will be favored. In this particular study, French has more types of interrogative constructions, including one with the *wh*-word *in situ*, and the derivationally simpler construction is transferred to Dutch, although it is not grammatical in that language.

Further evidence for cross-linguistic influence over time is provided by a large study of bilingual children, aged between 6 and 10 (Sorace, Serratrice, Filiaci, and Baldo 2009). Bilingual children were tested on whether they knew which antecedents null and overt pronouns refer to in null-subject languages. When null pronouns appear in the subject position of embedded clause (see the empty set symbol $\emptyset$ in (18)), they typically refer to the main clause subject, and we say it is an already mentioned Topic. When overt pronouns appear in the same position, they indicate that the Topic has

shifted, i.e., the previously mentioned (main clause) subject is no longer the pronoun referent. Note that the English translation of (18) is ambiguous while the Italian sentence has only one interpretation per pronoun. The null pronoun refers to the mother, while the overt pronoun can only mean that the daughter (or someone else), but not the mother, is putting on a coat.

(18)



La mamma dà un bacio alla figlia mentre **le**/ø si mette il capotto
 the mother gives a kiss to the daughter, while she REFL puts on the coat
 ‘The mother kisses her daughter, while she is putting on her coat.’

When they had to choose pronoun referents in a forced-choice task, the bilingual children made significantly more errors as compared to monolingual children. The fact that this was true for English–Italian as well as Italian–Spanish bilinguals¹² suggests that the two grammars in the mind of the bilingual affect how well the bilinguals can perform these subtle mappings between syntactic form to discourse function.

To be sure, other factors also influence which grammar of a bilingual affects the other. For example, language dominance (better knowledge of one language over the other) can cause transfer, as can the relative frequency of a structure in the two languages. Research on bilingual children is challenging because internal and external factors are difficult to control with actual children out there in the world. However, when these factors are controlled for and grammar interdependence is still attested, there is a strong indication that the linguistic factors of processing economy and processing complexity that we discussed are also at play.

In this section, I related the research consensus that even if children are acquiring two languages from birth, they are building two different grammatical systems that they are able to differentiate. Furthermore, we surveyed some studies pointing to the conclusion the two mental grammars interact in complex but linguistically predictable ways. These factors will remain important in the grammar-building process of adult learners, too.

¹² Recall that both Italian and Spanish are null-subject languages, hence have the same null and overt pronoun referents.

5.6 Exercises

Exercise 5.1. We concluded in the section on bilingual first language acquisition that building two grammars at the same time is possible and is frequently accomplished by bilingual children. However, even in simultaneous bilingual acquisition, the two grammars interact and influence each other. Discuss what this conclusion implies for the Critical Period Hypothesis and the frequent comparison of adult L2 learners with monolingual native speakers. (Hint: should we compare adult L2 learners with monolinguals. Why or why not?)

Exercise 5.2. Please see Table 5.3 below based on Brown (1973). The morphemes are arranged in a developmental sequence: if a child knows the morphemes at the bottom of the table, she certainly knows the ones at the top. A child producing the regular third person morpheme *-s* should already be accurate with regular plural. Check whether this is true in Alison's and Kathryn's speech samples below the table.

Table 5.3 Developmental sequence of morpheme acquisition, following Brown (1973)

Morpheme	Example	Age of Mastery* (in Months)
Present Progressive	<i>-ing: Mommy driving</i>	19–28
<i>In</i>	<i>Ball in cup</i>	27–30
<i>On</i>	<i>Doggie on sofa</i>	27–33
Regular plural	<i>-s: Kitties eat my ice cream.</i> Forms: /s/, /z/ and /ɪz/ <i>cats, dogs, classes, wishes</i>	27–33
Irregular past	<i>came, fell, broke, sat, went</i>	25–46
Possessive	<i>'s: Mommy's balloon broke</i> Forms: /s/, /ɪz/ and /ɪz/ as in regular plural	26–40
Uncontractible copula (Verb to be as main verb)	<i>He is.</i> (Response to “Who is sick?”)	28–46
Articles	<i>I see a kitty.</i>	28–46
Regular past	<i>-ed: Mommy pulled the wagon</i> Forms: /d/, /t/, /ɪd/ <i>pulled, walked, glided</i>	26–48
Regular third person	<i>-s: Kathy hits</i> Forms: /s/, /z/, and /ɪz/	28–50
Irregular third person	<i>Does, has</i>	28–50

Uncontractible auxiliary	<i>He is.</i> (Response to “Who is wearing your hat?”)	29–48
Contractible copula	<i>Man’s big</i> <i>Man is big</i>	29–49
Contractible auxiliary	<i>Daddy’s eating</i> <i>Daddy is eating</i>	30–50

* Used correctly 90% of the time in obligatory contexts. Adapted from Bellugi and Brown (1964); R. Brown (1973); and J. Miller (1981).

ALISON

Age 1;4

Child: (picks up toy) cow. cow. cow.

Child: (tries to stand it on chair) chair. chair.

Mother: (child gives cow to mother for help) what, darling?

Child: Mama.

Age 1;4

Child: (offers biscuit to mother) Mommy

Mother: Oh, thank you.

Child: (looks at cup) juice.

Mother: Shall we have some juice?

Child: (looks in cup) cup. (picks out two cups and holds one cup out to mother) Mommy. juice.

Mother: Mommy juice.

Child: No. (puts down one cup) Baby!

Mother: Baby.

Child: (picks up another cup) Mommy. Juice.

Age 1;9

Child: (wearing a jacket and pointing to her neck) up. up.

Mother: What?

Child: Neck. up.

Mother: Neck? what do you want? what?

Child: Neck.

Mother: What’s on your neck?

Child: (points to zipper and lifts her chin up) zip. zip. zip.

Age 2;4

Child: (Mother is wearing a microphone) I want to try that – that necklace.

Mother: You want to try this necklace? Okay. you know what this is called?

Child: Something.

Mother: See, it’s a microphone. there. (puts the microphone on Alison)

Child: I don't want it.

Mother: You don't want it?

Child: No, you want it on, you have it on.

Age 2;4

Child: (climbing on chair) I'm gonna sit on her, may I? I wanna sit on here.

Child: (later) oh, I don't want drink it out cup. I want drink it out can.

Mother: Oh, what did I say about that? what did I say about drinking it out of the can?

Child: Mommy, I want it.

Mother: You want it?

Child: I want drink it out can.

Age 2;10

Mother: (a baby boy is in the room) oh, maybe he'd like to see the truck?

Child: He – he may want to play with the truck.

Mother: Okay.

Child: Maybe he'll play with the truck. he can play with the truck. I think he'll play with the truck.

Mother: Yes, I think he likes the truck.

KATHRYN

Age 1;9

Child: (picks up red bean bag shaped like a frog) Santa Claus.

Mother: Santa Claus? that's a frog, honey, that's not Santa Claus. that's a frog. red frog.

Child: Frog. (puts frog on car) sits.

Mother: Yes, he is sitting down. that's right.

Age 2;9

Child: (looking under her skirt) I just have see

A: (language researcher has come to visit. The child hasn't seen her for six weeks) hi.

Child: Hi.

A: Hi, Kathryn. I haven't seen you in such long time.

Child: You came. you come a lot of days.

A: I do come a lot.

Child: You come lots of weeks. again and again and again.

A: I didn't come last week.

Child: So you came this week.

Age 2;11

Child: (playing ‘airplane trip’ with the language researcher) mow, why don’t we shut the doors?

A: Hmmm?

Child: Why don’t you shut your – shut your door? shut so no air can come in.

A: I did.

Child: Let’s get. ssh! ssh! and some nurses bring some – some food.

A: Who?

Child: The – the – the – the – lady who lives in the airplane.

A: Who is the lady who lives in the airplane? what did you call her?

Child: A lady

A: Did you call her a nurse or a stewardess?

Child: A stewardess.

Data from Bloom, L. and Lahey, M. (1978). *Language Development and Language Disorders*. New York: Wiley.

Exercise 5.3. Using Anne’s production data from the CHILDES database, provide an MLU calculation in morphemes for two of the files from the beginning and the end of data collection. In the first file, Anne is 1 year 10 months old (1;10) and in the second file she is 2;9. You can retrieve the files from the CHILDES database using the link below. Anne’s data is part of the Manchester corpus. To get to the original files you have to open this link, then download the folder called Manchester.zip and unzip it. All Anne’s files are there: open the files called “anne01a.cha” and “anne34a.cha” using a word processing application such as Microsoft Word.

<http://childes.psy.cmu.edu/data/Eng-UK-MOR/http://childes.psy.cmu.edu/data/Eng-UK-MOR/>

This is how to calculate MLU:

Exclude from your count: imitations, elliptical answers, unintelligible utterances, false starts and reformulations within utterances, noises, discourse markers, identical utterances.

Count as one morpheme: uninflected lexical morphemes, contractions, inseparable linguistic units, irregular past tense, plurals which do not occur in singular form.

Example: *The boy’s father slowly filled the glass with juice.*

The	boy	's	father	slow	-ly	fill	-ed	the	glass	with	juice
1	2	3	4	5	6	7	8	9	10	11	12

This is one sentence containing 9 words and 12 morphemes. You can find some of the conventions of the transcripts here: <http://chilides.psy.cmu.edu/manuals/CHAT.pdf>.

Exercise 5.4. Here are some more examples of errors children make in complex constructions.

- A. Children close to five years of age were not at all capable of correctly repeating relative clause constructions with genitives such as the one below (Frizelle and Fletcher, 2013).

There is the girl **whose toy** Anne broke in the garden.

- B. At 4;1, a child produced erroneous sentences such as these (Guasti, Thornton, and Wexler 1995).
- What did he didn't wanna bring to school?
 - What did the spaceman didn't like?
- C. English-speaking kids aged 4;1–5;4 produce the following in an elicitation task (Thornton 1990):
- What do you think which animal says “woof woof”?
 - What do you think which Smurf really has roller skates?
 - What do you think what Cookie Monster eats?
 - How do you think how Superman fixed the car?
- D. 12 English-speaking kids aged 4;5–6 produced the following question where the adult target *wh*-word is *whose* (Gavrusseva and Thornton 2001).
- Who do you think's flower fell off?
 - Who do you think's sunglasses Pocahontas tried on?
 - Who do you think's Spiderman saved cat?

The construction in C is called medial *wh* and is an attested option in languages such as German. The construction in D may also be analyzed along the same lines. Think about the elements of the sentence that the children repeat. In which syntactic positions do you find them? What do these further examples mean in light of our discussion of structural and derivational complexity as factors in acquisition? In each case, explain exactly where the complexity lies, and what the children are doing to ameliorate it.